

LEONARD COURT

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EDUCATION

ETH Zurich, Switzerland	Expected Aug 2028
Master of Science in Robotics, Systems and Control	
University of Illinois at Urbana-Champaign (Uofl)	2025 - 2026
Study Abroad Program in Mechanical Engineering	
École Polytechnique Fédérale de Lausanne (EPFL), Switzerland	2023 - 2026
Bachelor of Science in Mechanical Engineering	GPA : 5.5/6

Related coursework :

Robot Dynamics	Probabilistic Artificial Intelligence	Model Predictive Control	Vision Algorithm	DPOC
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TECHNICAL SKILLS

Programming Languages : C++, C, Python, Matlab

Softwares/Tools : PyTorch, ROS 2, Docker, Eigen, Solidworks, Fusion 360, Abaqus, Catia

Spoken languages : French C2(Native), English C1 (Advanced), Spanish B1 (Intermediate)

WORK EXPERIENCE

Ecole Polytechnique Fédérale de Lausanne (EPFL), Switzerland	Aug 2024 - Jul 2025
Teaching Assistant	

- Supported learning for cohort of 500+ students by explaining key concepts and problem-solving methods
- Guided students through exercises and provided strategies for academic success, graded mid-term tests and offered feedback
- Cours : *Physics: Mechanics (PHYS 101)* (Prof. Philippe Müllhaupt), *Physics: Thermodynamics (PHYS 106)* (Prof. Sylvain Bréchet), *Physics: Thermodynamics (PHYS 106)*(Prof. Jérémy Genoud), *Information, Computation, Communication (CS 119)*(Prof. Olivier Lévéque)

PROJECT HIGHLIGHTS

Autonomous Navigation Stack in Simulation — C++17, Python, PyTorch, ROS 2, Eigen, Docker	Sep 2024 - Present
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- Developed an end-to-end autonomous navigation stack in simulation, combining LiDAR-based localization (EKF, particle filter), sensor fusion, A* path planning, and LQR trajectory tracking.
- Generated and evaluated navigation scenarios through 100+ Monte Carlo simulations under varying sensor noise, measuring localization accuracy, collision rate, tracking error, and computational cost.
- Trained a PyTorch CNN to imitate A*-based path generation, benchmarking the learned planner against the classical approach for success rate, path efficiency, and inference time; packaged the system with ROS 2, Docker, unit tests, and CI

EPFL Rocket Team – Firehorn 2025 (Ground Station Subsystem Engineer)	Sep 2024 - Jul 2026
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- Contributed to mechanical design, prototyping, and testing of ground support components for competition-level rocketry
- Developed CAD models and manufactured prototype parts using SolidWorks, 3D printing, CNC machining, and assembly
- Supported system integration between mechanical, electrical, and operational subsystems through testing and troubleshooting
- Collaborated in a multidisciplinary engineering team, applying problem-solving and design iteration to deliver practical solutions

Candy Crush (C Programming Project)

- Developed the full Candy Crush game in C and implemented AI capable of playing autonomously

School support platform, 'The Academy'	Jan 2021 - Jul 2023
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- Created and led online academic support platform bringing together several thousand students
- Organized peer-to-peer learning opportunities and student-led classes
- Collaborated with private advertisers, generating revenue through marketing partnerships

INTERESTS, ACTIVITIES & LEADERSHIP

EPFL Coaching : Student integration & mentoring	Sep 2024 - Jul 2025
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Sports : Practiced gymnastics at national level, Surf, Kite-Surf, Wing Foil, Wind-Surf